WORKED SOLUTION

STEP II 2021, Question 6

Pure • Circle Geometry Approximations

STEP II 2021, Question 6

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

Everything happens in one triangle: A, the centre O of the road, and the centre O' of the cyclist's circle. Symmetry puts O' on the line through O that bisects $\angle AOB$, and the tangency fixes the distance OO'. The cosine rule and the sine rule in that triangle then give d and $\alpha - \theta$. For the last part, *much less than 1* means we keep first-order terms in w/R and use small-angle approximations.

Solution

Take O as origin with the line of symmetry vertical, so that $A = (-(R+w)\sin\alpha,\ (R+w)\cos\alpha)$ and B is its mirror image. The road is the region $R \le r \le R+w$, where r is the distance from O.

Part (i)

The point of the chord AB closest to O is its midpoint M, since $OM \perp AB$ (triangle OAB is isosceles). Then

$$OM = (R+w)\cos\alpha \le \tfrac{1}{2}(R+w) < \tfrac{1}{2}(R+R) = R,$$

using $\cos\alpha \le \frac{1}{2}$ (because $\frac{1}{3}\pi \le \alpha \le \frac{1}{2}\pi$) and then $w < R$. So M lies strictly inside the inner circle, which is off the road. Any straight ride from A to B passes through M, so it cannot stay on the road.

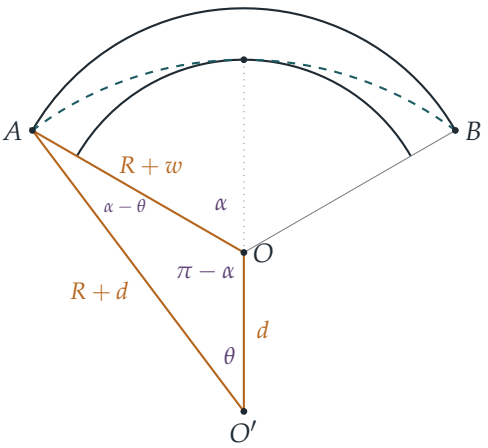
Part (ii)

Where is O'? The circle passes through A and B, so its centre is equidistant from them and lies on the perpendicular bisector of AB, the line of symmetry through O. It touches the inner circle and is bigger than it ($R + d > R$), so the inner circle sits inside it, touching it internally. For internally tangent circles the distance between centres is

the difference of the radii:

$$OO' = (R + d) - R = d.$$

The path bulges outwards, towards the road, so O' lies on the opposite side of O from the road, as in the figure. In triangle AOO' we then have $OA = R + w$, $OO' = d$, $O'A = R + d$, and the angle at O is $\pi - \alpha$ (it is supplementary to $\angle AOM = \alpha$). The angle at O' is $\frac{1}{2}\angle AO'B = \theta$, so the angle at A is $\pi - (\pi - \alpha) - \theta = \alpha - \theta$.



(a) The cosine rule in triangle AOO', with $\cos(\pi - \alpha) = -\cos\alpha$, gives

$$\begin{aligned}(R + d)^2 &= (R + w)^2 + d^2 + 2(R + w)d\cos\alpha \\ R^2 + 2Rd &= R^2 + 2Rw + w^2 + 2(R + w)d\cos\alpha \\ 2d\big(R - (R + w)\cos\alpha\big) &= w(2R + w),\end{aligned}$$

so

$$d = \frac{w(2R + w)}{2\big(R - (R + w)\cos\alpha\big)}.$$

The bracket in the denominator is positive by part (i), so this is a sensible positive length.

(b) The sine rule in the same triangle, pairing the angle $\alpha - \theta$ (opposite $OO' = d$) with the angle $\pi - \alpha$ (opposite $O'A = R + d$), gives

$$\frac{\sin(\alpha - \theta)}{d} = \frac{\sin(\pi - \alpha)}{R + d} \quad \Longrightarrow \quad \sin(\alpha - \theta) = \frac{d\sin\alpha}{R + d}.$$

Part (iii)

Write $\varepsilon = w/R$, so $0 < \varepsilon \ll 1$. Dividing the result of (a) by R,

$$\frac{d}{R} = \frac{\varepsilon(2 + \varepsilon)}{2\big(1 - (1 + \varepsilon)\cos\alpha\big)}.$$

Since $\cos\alpha \le \frac{1}{2}$, the denominator is at least $2\big(1-\frac{1}{2}(1+\varepsilon)\big) = 1-\varepsilon$, so

$$0 < \frac{d}{R} \le \frac{\varepsilon(2+\varepsilon)}{1-\varepsilon} \approx 2\varepsilon,$$

which is much less than 1. (To first order, $d/R \approx \varepsilon/(1-\cos\alpha)$.)

Ask yourself A small sine only tells us that the angle is close to 0 *or* close to π. Which is it?

From (b), $0 < \sin(\alpha-\theta) = \dfrac{(d/R)\sin\alpha}{1+d/R} < \dfrac{d}{R}$, which is small. The angle $\alpha-\theta$ is an angle of triangle AOO', so it is positive, and it is less than $\alpha \le \frac{1}{2}\pi$ because $\theta > 0$. On $(0, \frac{1}{2}\pi)$ the sine is increasing with $\sin x \ge \frac{2}{\pi}x$, so $\alpha-\theta \le \frac{\pi}{2}\sin(\alpha-\theta) < \frac{\pi}{2}\cdot\frac{d}{R}$, which is also much less than 1.

Part (iv)

My friend rides along the outer edge, an arc of radius $R+w$ subtending 2α at O, so their path has length $L = 2\alpha(R+w)$. My path is the arc of radius $R+d$ on the far side of AB from O'; it subtends 2θ at O', so its length is $2\theta(R+d)$. Therefore

$$\begin{aligned} S &= 2\alpha(R+w) - 2\theta(R+d) \\ &= 2(R+d)(\alpha-\theta) + 2\alpha(R+w) - 2\alpha(R+d) \\ &= 2(R+d)(\alpha-\theta) + 2\alpha(w-d), \end{aligned}$$

as required (we added and subtracted $2\alpha(R+d)$).

The approximation. We now work to first order in ε, ignoring terms of order ε^2 relative to those we keep. By (iii), $\alpha-\theta$ is small, so $\alpha-\theta \approx \sin(\alpha-\theta)$, with error of order $(\alpha-\theta)^3$. By (b),

$$2(R+d)(\alpha-\theta) \approx 2(R+d)\cdot\frac{d\sin\alpha}{R+d} = 2d\sin\alpha.$$

Hence

$$S \approx 2d\sin\alpha + 2\alpha w - 2\alpha d = 2\alpha w - 2d(\alpha-\sin\alpha).$$

Now use $d \approx \dfrac{w}{1-\cos\alpha}$ from (iii):

$$S \approx 2w\left(\alpha - \frac{\alpha-\sin\alpha}{1-\cos\alpha}\right) = 2w\cdot\frac{\alpha-\alpha\cos\alpha-\alpha+\sin\alpha}{1-\cos\alpha} = \frac{2w(\sin\alpha-\alpha\cos\alpha)}{1-\cos\alpha}.$$

Finally $L = 2\alpha R(1+\varepsilon) \approx 2\alpha R$, and since S is already of first order in ε, replacing L by $2\alpha R$ only changes S/L at second order. So

$$\frac{S}{L} \approx \left(\frac{\sin\alpha-\alpha\cos\alpha}{\alpha(1-\cos\alpha)}\right)\frac{w}{R},$$

which is the required fraction.

Checks

- At $\alpha = \frac{1}{2}\pi$ the fraction is $\frac{2}{\pi} \cdot \frac{w}{R} \approx 0.637\,\frac{w}{R}$; at $\alpha = \frac{1}{3}\pi$ it is about $0.654\,\frac{w}{R}$. Both are positive, as they must be, since $\tan\alpha > \alpha$ gives $\sin\alpha > \alpha\cos\alpha$.
- With $R = 1$, $w = 0.01$, $\alpha = \frac{1}{2}\pi$: the exact formulas give $d \approx 0.01005$, $\alpha - \theta \approx 0.00995$ and $S/L \approx 0.006285$, against the approximation 0.006366. The relative error, about 1%, is of order w/R, as it should be, and it shrinks tenfold when w is divided by ten.
- A computer check placed A and B in coordinates, found the tangent circle directly, and confirmed the formulas for d, $\sin(\alpha - \theta)$ and S for thousands of random choices of R, w and α.

Common slips

- Taking $OO' = R + d + R$ or $OO' = w$: the circles touch *internally*, so the distance between centres is the difference of the radii, d.
- Using $\cos\alpha$ instead of $\cos(\pi - \alpha)$ for the angle at O in triangle AOO', which flips the sign of the last term.
- In (iii), saying only that $\sin(\alpha - \theta)$ is small: you also need $\alpha - \theta$ to lie in $(0, \frac{1}{2}\pi)$ before you can conclude that the angle itself is small.

Beyond STEP

"Much less than 1" is the language of perturbation theory: expand in a small parameter ε, keep the leading term, and make sure that the error really is of higher order. At university this is made precise with Taylor's theorem and big-O notation, and here it would show that S/L equals the given fraction times ε, plus a term of order ε^2. For a careful treatment of Taylor's theorem with remainder, see Abbott, *Understanding Analysis*; for more problems in the same spirit, see Siklos, *Advanced Problems in Mathematics*.

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